

Themes

Exercise 5 — ISS dataviz, Spring 2026

YOUR NAME HERE

Invalid Date

Table of contents

Task 1: Session check-in	1
Task 2: Cheese!	1
Data description and cleaning	1
Base plot	3
Recreate nice plot	4
Task 3: Extension Ugly plot	5
Bibliography	5

Task 1: Session check-in

Three interesting or exciting things:

1. Something
2. Something
3. Something

Three muddy or unclear things:

1. Something
2. Something
3. Something

Task 2: Cheese!

Data description and cleaning

This data comes from cheese.com, and it was a #TidyTuesday dataset in June 2024

There are 19 columns, and they're documented here—make sure you consult that page!

The code below loads and cleans the data, resulting in a data frame named `cheeses_milk_country` that contains the number of cheeses made by the top 8 cheese-producing countries, across three types of milk. **You don't need to modify anything in this chunk of code—you only need to run it.**

```

library(tidyverse)

# Load the full data
cheeses_raw <- read_csv("data/cheeses.csv")

# The milk and country columns contain multiple values, like "cow, goat" and
# "Italy, France". separate_longer_delim() splits those into separate rows
# based on the ",", so that "cow, goat" would turn into one row for cow and
# one
# for goat, and so on
cheeses <- cheeses_raw |>
  separate_longer_delim(milk, delim = ", ") |>
  separate_longer_delim(country, delim = ", ") |>
  # Technically England, Scotland, GB, and the UK are all different things,
  # but
  # for the sake of visualization, we can lump them all together
  mutate(country = case_match(country,
    "England" ~ "United Kingdom",
    "Great Britain" ~ "United Kingdom",
    "Scotland" ~ "United Kingdom",
    .default = country
  ))

# Find the most common milk types
top_milk_types <- cheeses |>
  group_by(milk) |>
  summarize(total = n()) |>
  filter(total > 100)

# Find the most common countries
top_countries <- cheeses |>
  group_by(country) |>
  summarize(total = n()) |>
  filter(total > 30)

# Calculate the number of milk types across the most common countries
cheeses_milk_country <- cheeses |>
  filter(milk %in% top_milk_types$milk, country %in% top_countries$country) |
  >
  group_by(milk, country) |>
  summarize(total = n()) |>
  ungroup() |>
  # Capitalize the milk types
  mutate(milk = str_to_title(milk)) |>
  # Sort by country and milk type and lock in the order for better plotting
  arrange(desc(total), milk) |>
  mutate(across(c(country, milk), \(x) fct_inorder(x)))
cheeses_milk_country

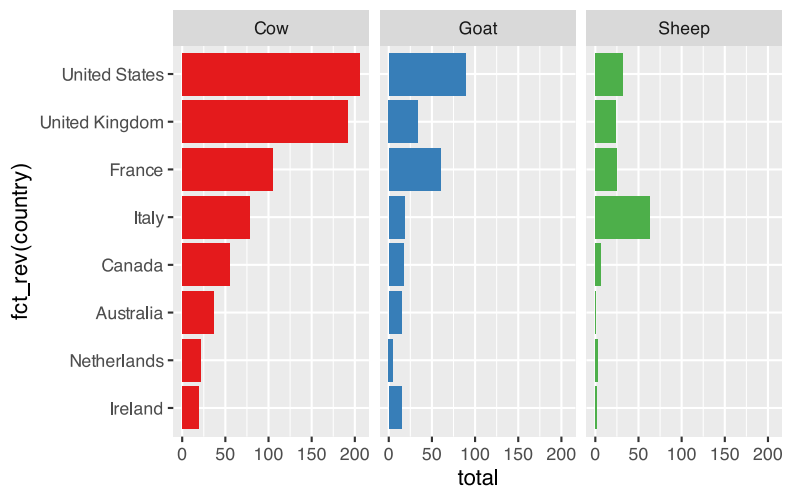
```

```
# A tibble: 24 × 3
  milk country    total
  <fct> <fct>      <int>
1 Cow   United States  206
2 Cow   United Kingdom 192
3 Cow   France       105
4 Goat  United States   89
5 Cow   Italy          78
6 Sheep Italy          63
7 Goat  France          60
8 Cow   Canada          55
9 Cow   Australia        36
10 Goat United Kingdom  34
# i 14 more rows
```

Base plot

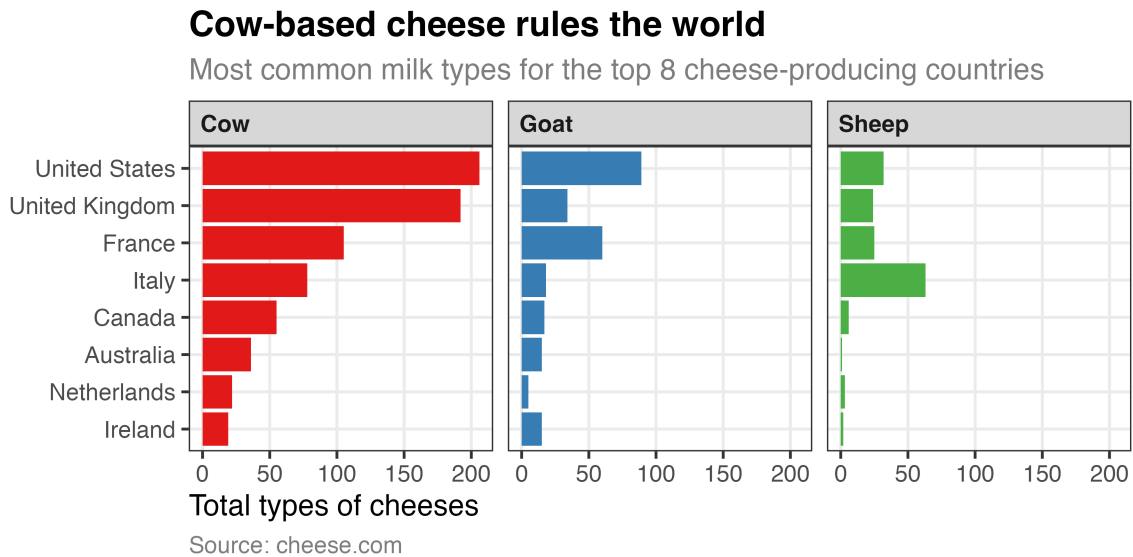
Here's a basic bar chart showing the counts of countries and milk types. It's stored as an object named `base_plot`:

```
base_plot <- ggplot(
  cheeses_milk_country,
  aes(x = total, y = fct_rev(country), fill = milk)
) +
  geom_col() +
  guides(fill = "none") +
  # These colors come from ColorBrewer 2:
  # https://colorbrewer2.org/#type=qualitative&scheme=Set1&n=3
  scale_fill_brewer(palette = "Set1") +
  facet_wrap(vars(milk))
base_plot
```



Recreate nice plot

Run the code chunk below to see a plot.



Your job is to use `labs()` and `theme()` to recreate this plot:

Some helpful big hints:

- The image is 6 inches wide and 3 inches tall. Use chunk options like `fig-width` and `fig-height` to control the dimensions (see <https://datavizsp26.classes.andrewheiss.com/resource/quarto.html#chunk-options>)
- Use `labs()` to modify the following things: `x`, `y`, `title`, `subtitle`, and `caption`
- Use `theme_bw()`
- Use `theme()` to adjust these things:
 - Strip text
 - Axis title
 - Panel grid major
 - Plot title
 - Plot subtitle
 - Plot caption

In the end, your code will look something like this:

```
base_plot +  
  labs(x = "BLAH", title = "BLAH") +  
  theme_bw() +  
  theme(  
    strip.text = element_text(BLAH),  
    axis.title = element_text(BLAH)  
  )
```

MAKE SURE you use the theme resources and cheatsheets from the content page. Also consider using `{ggthemeassist}` to help with the different settings.

```
# Do stuff here
```

Task 3: ~~Extension~~ Ugly plot

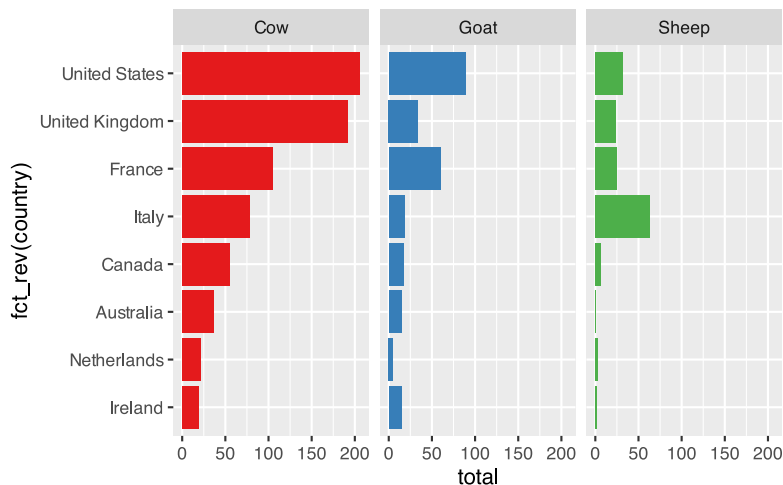
For your extension, forget all the wonderful CRAP design principles you just learned and try your hardest to **make the ugliest plot in the world**. Take `base_plot` and create the worst, ugliest, least CRAPful plot ever.

Change the colors, modify the theme, add labels, etc. and make this ugly. You can leave the `geom_col()` as is, or you can mess with it too (or even change it to a pie chart or heatmap or something if you feel up to it).

This is your chance to play around and explore.

```
# Do stuff here
```

```
base_plot +  
  theme()
```



Bibliography